



Dr. N.G.P. INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, New Delhi & Affiliated to Anna University, Chennai)
Recognized by UGC & Accredited by NAAC with A+ and NBA (BME, CSE, ECE, EEE and Mech)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Students name: KEVIN K R MAHASMRTI S S	Class : III CSE A
Guide Name : Ms. P. Nisha AP/CSE	Domain: Machine Learning in Agriculture
Predictive Hybridization of Orphan Crops using Machine Learning Models	

Abstract :

This project presents a machine learning-based system for predicting hybrid outcomes in orphan crops—underutilized yet vital food sources known for their climate resilience and nutritional value. Traditional hybridization in plant breeding is a time-intensive and resource-demanding process. By applying predictive modeling techniques, this project aims to forecast the success of specific cross-breeding combinations based on parental crop characteristics such as yield potential, disease resistance, and drought tolerance. The system is designed to assist crop scientists and agricultural planners in making faster and more accurate decisions, especially in resource-constrained settings. This work promotes efficient hybrid selection, reduces the breeding cycle duration, and contributes to improved food security by empowering research on neglected crop species. The project also includes a simple UI for trait input and prediction display, making it accessible to users with non-technical backgrounds.

Objectives:

- To develop a machine learning model that predicts the success of crop hybrid combinations.
- To utilize trait-based data from parental crops for training and testing the model.
- To support orphan crop research through faster and more informed hybrid selection.
- To create a user interface for trait input and prediction output.

Materials and Methods: (Proposed Work Flow)

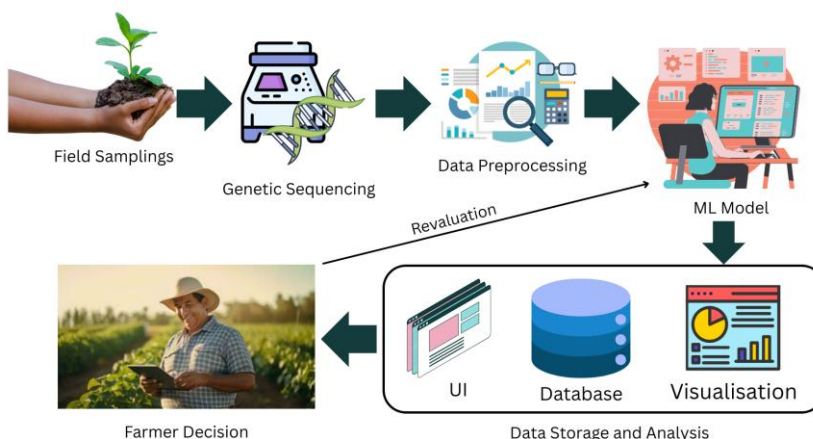
• **Data Collection:** Use open-source or synthetic data on orphan crops, including traits like height, yield, resistance, and climate adaptation.

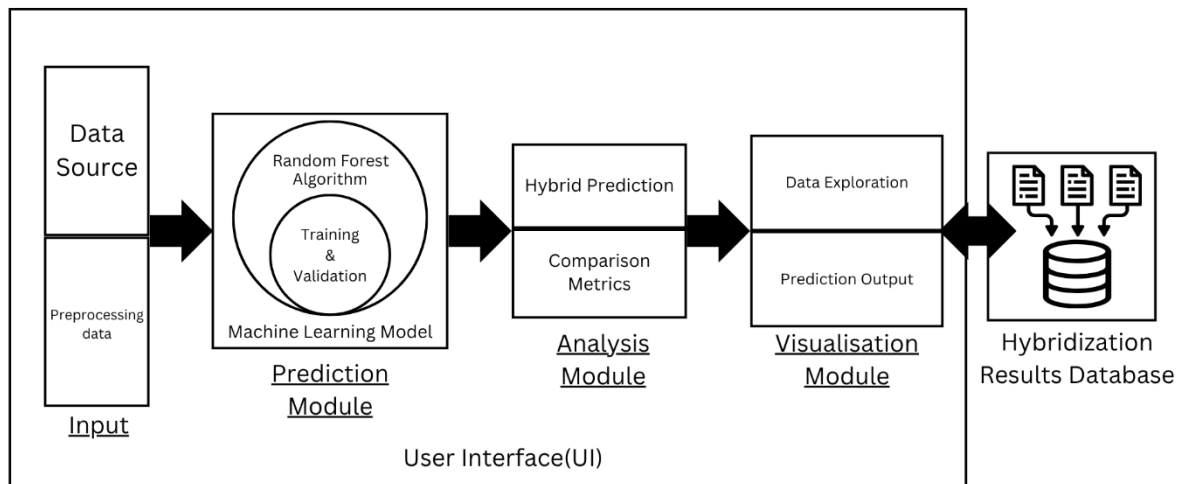
• **Preprocessing:** Feature encoding, normalization, and handling missing values.

• **Model Selection:** Train and test classification algorithms (Random Forest).

• **Prediction:** Output whether a particular hybrid is likely to be successful or not.

• **Evaluation:** Use metrics like accuracy, precision, recall, and F1-score to validate the model.

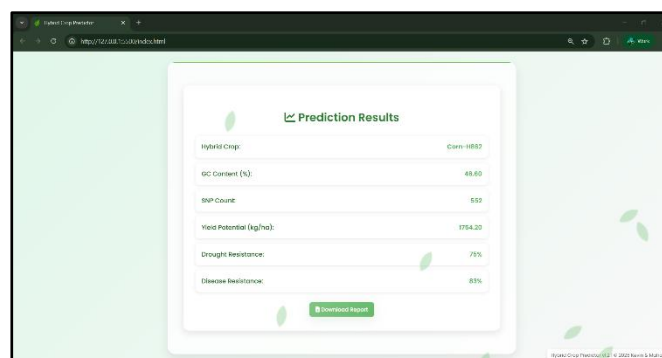
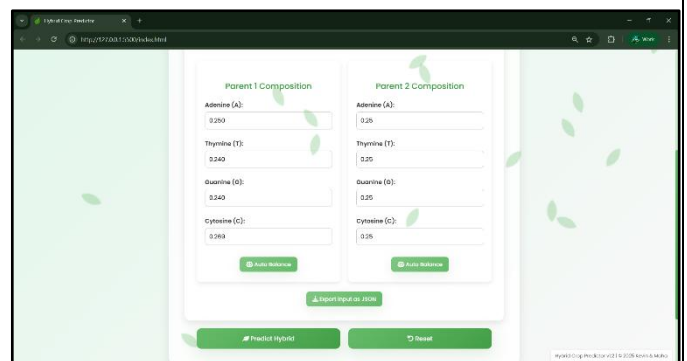
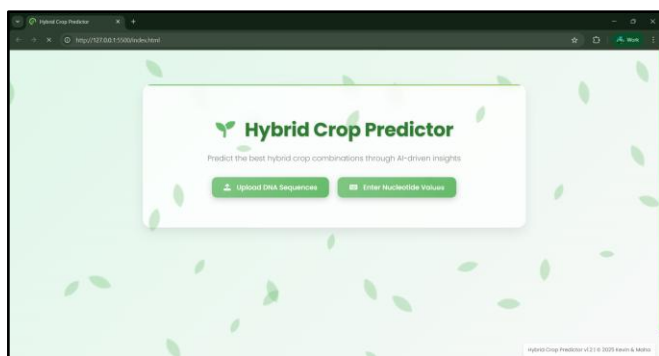




Block Diagram

Results:

Model performance and prediction samples will be displayed after training with a finalized dataset. Initial simulations show promising direction in classifying potential hybrid success based on trait-based features.



Conclusion : The project demonstrates the potential of machine learning to assist in predictive hybridization for orphan crops. By enabling faster, data-driven decisions in selecting parent combinations, the system offers a scalable solution to challenges in agricultural research and crop improvement. It provides a technological aid to reduce experimentation cycles and promotes the utilization of locally important but globally neglected crops.

Achievements:

- Selected as one of the innovative projects under the department's Project-Based Learning (PBL) initiative.
- Shortlisted for presentation during internal project expo held in Semester IV.
- Appreciated by faculty panel for addressing food security through data-driven crop improvement.
- Demonstrated significant improvement in prediction accuracy during internal evaluation using test datasets.
- Positive feedback received for project scope and practical applicability during NBA pre-evaluation sessions.
- Continual semester-wise development plan approved for future academic integration.

References:

1. Varshney, R. K., et al. (2021). Orphan Crops in the Genomics Era: Challenges and Opportunities. *Trends in Plant Science*.
2. Subrahmanyam, P., et al. (2015). Hybrid Breeding Approaches in Underutilized Crops. *Indian Journal of Genetics*.
3. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*.
4. CGIAR (2020). Open access datasets on orphan crops. www.cgiar.org